

1.What is the bias-variance tradeoff?

- The bias-variance tradeoff is the balance between underfitting and overfitting in a machine learning model.
- High bias means the model is too simple and leads to underfitting.
- High variance means the model is too complex and leads to overfitting.
- The goal is to find the right balance so the model performs well on both training and unseen data.
- A good model has low bias and low variance, resulting in accurate predictions and good generalization.

2.What is L1 vs L2 regularization?

L1 Regularization (Lasso): L1 Regularization is a technique used to prevent overfitting by adding the absolute values of the model coefficients to the loss function. It reduces the importance of less useful features and can make some coefficients exactly zero, effectively performing feature selection. This helps create a simpler and more interpretable model.

L2 Regularization (Ridge): L2 Regularization is a technique that prevents overfitting by adding the squares of the model coefficients to the loss function. It reduces the magnitude of all coefficients but usually does not make them zero. This helps the model generalize better to new data while keeping all features in the model.

3.How does gradient descent work?

Gradient Descent is an optimization algorithm used to minimize the loss (error) function by updating the model's parameters iteratively. It moves in the direction of the negative gradient, where the error decreases the fastest. The process continues until the model reaches the minimum error and achieves the best possible performance.

4.What is the difference between Batch, Stochastic, and Mini-batch Gradient Descent?

Batch Gradient Descent updates the model using the entire training dataset in each iteration, making it accurate but slow. Stochastic Gradient Descent updates the model

using one training sample at a time, making it faster but less stable. Mini-batch Gradient Descent uses a small batch of training samples, providing a good balance between speed and accuracy.

5.What is overfitting and how do you prevent it?

Overfitting occurs when a machine learning model learns the training data too closely, including its noise, resulting in poor performance on new or unseen data. It can be prevented by using techniques such as regularization (L1/L2), cross-validation, dropout, early stopping, collecting more training data, or reducing model complexity. A well-generalized model performs well on both training and testing data.

6.What is the role of activation functions in neural networks?

Activation functions help a neural network learn

n complex patterns by deciding whether a neuron should be activated or not. They allow the network to make better predictions by introducing non-linearity into the model.

7.Why is ReLU preferred over Sigmoid?

ReLU is preferred over Sigmoid because it is faster to compute and helps the neural network learn more efficiently. It also reduces the vanishing gradient problem, making it better for training deep neural networks.

8.What is backpropagation?

Backpropagation is the process of updating the weights of a neural network by calculating the error from the output layer back to the input layer. It helps the model reduce prediction errors and improve its accuracy during training

9.What causes the vanishing gradient problem?

The vanishing gradient problem occurs when gradients become very small during backpropagation, especially in deep neural networks. As a result, the early layers learn very slowly or stop learning, reducing the model's performance.

10.What is the purpose of Batch Normalization?

Batch Normalization is used to normalize the inputs of each layer during training. It helps the model train faster, improves stability, and reduces the chances of overfitting.

11.What is the core difference between CNNs and RNNs?

CNNs (Convolutional Neural Networks) are mainly used for image and video data, while RNNs (Recurrent Neural Networks) are used for sequential data like text, speech, and time series. CNNs capture spatial features, whereas RNNs capture information from previous inputs in a sequence.

12.How do Transformers differ from RNNs?

Transformers process all input data at the same time, making them faster and better for handling long sequences. RNNs process data one step at a time, making them slower and less effective for long-term dependencies.

13.What is Dropout?

Dropout is a technique used to prevent overfitting in neural networks by randomly turning off some neurons during training. This helps the model learn better and improves its ability to generalize to new data

14.What is the difference between Precision and Recall?

Precision measures how many of the predicted positive cases are actually correct, while Recall measures how many of the actual positive cases the model correctly identifies. High precision reduces false positives, whereas high recall reduces false negatives.

15.What is Transfer Learning?

Transfer Learning is a technique where a pre-trained model is reused for a new but similar task. It reduces training time, requires less data, and often improves model performance.

